

OUTSIDE THE DATABASE (Non-Relational Assets)

Embedded Hardware Module

ESP32-C3 RISC-V + MAX30102 Optical PPG (3.3V, I2C GPIO4/5 @ 400kHz)

Machine Learning Static Weights

assets/ml/weights.json (Tiny MLP 5->16->8->1, 241 params in Dart Isolate)

Over-The-Air BLE GATT Stream

12-Byte Little-Endian Heart Rate Notification Packet (1 Hz, protocol v1)

Generated PDF Biomonitoring Reports

Rendered offline on device via PdfReportGenerator into app storage directory

Architectural Distinction & Data Boundary Notes

- These entities are NOT SQLite tables and are NOT stored as relational records.
- Hardware & BLE streaming supply transient raw samples ingested into the `samples` table.
- ML weights are compiled assets; model outputs populate the `ai_results` table.
- PDF reports aggregate data from `sessions`, `session_stats`, and `ai_results` on demand.